

◆ Equations of motion

Equations that can be used to make calculations about objects that are moving with uniform acceleration.

Equations of motion for uniformly accelerated motion

SYLLABUS CONTENT

- The **equations of motion** for solving problems with uniformly accelerated motion as given by:

$$s = \frac{(u + v)}{2}t$$

$$v = u + at$$

$$s = ut + \frac{1}{2}at^2$$

$$v^2 = u^2 + 2as$$

The five quantities u , v , a , s and t are all that is needed to fully describe the motion of an object that is moving with *uniform* acceleration.

- u = velocity (speed) at the start of time t
- v = velocity (speed) at the end of time t
- a = acceleration (constant)
- s = displacement occurring in time t
- t = time taken for velocity (speed) to change from u to v and to travel a distance s .

If any three of the quantities are known, the other two can be calculated using the first two equations highlighted below.

If we know the initial velocity u and the uniform acceleration a of an object, then we can determine its final velocity v after a time t by rearranging the equation used to define acceleration:

$$a = \frac{(v - u)}{t}$$

This gives:



$$v = u + at$$

If an object moving with velocity u accelerates uniformly to a velocity v , then its average velocity is:

$$\frac{(u + v)}{2}$$

Then, since distance = average velocity $\times$ time:

$$s = \frac{(u + v)}{2}t$$



These two equations can be combined mathematically to give two further equations, shown below. These very useful equations do not involve any further physics theory, they just express the same physics principles in a different way.

$$s = ut + \frac{1}{2}at^2$$

$$v^2 = u^2 + 2as$$



LINKING QUESTION

- How are the equations for rotational motion related to those for linear motion?

This question links to understandings in Topic A.4.

WORKED EXAMPLE A1.5

A Formula One racing car (see Figure A1.34) accelerates from rest at 18 m s^{-2} .



■ **Figure A1.34** Formula One racing cars at the starting grid

- a Calculate its speed after 3.0 s.
- b Calculate how far it travels in this time.
- c If it continues to accelerate at the same rate, determine its velocity after it has travelled 200 m from the start.

Answer

a $v = u + at = 0 + (18 \times 3.0) = 54 \text{ m s}^{-1}$

b $s = \frac{(u + v)}{2}t = \frac{(0 + 54)}{2} \times 3.0 = 81 \text{ m}$

But note that the distance can be calculated directly, without first calculating the final velocity, as follows:

$$s = ut + \frac{1}{2}at^2 = (0 \times 3.0) + (0.5 \times 18 \times 3.0^2) = 81 \text{ m}$$

c $v^2 = u^2 + 2as = 0^2 + (2 \times 18 \times 200) = 7200$
 $v = 85 \text{ m s}^{-1}$

WORKED EXAMPLE A1.6

A train travelling at 50 m s^{-1} (180 km h^{-1}) needs to decelerate uniformly so that it stops at a station 2.0 kilometres away.

- a Determine the necessary deceleration.
- b Calculate the time needed to stop the train.

Answer

a $v^2 = u^2 + 2as$
 $0^2 = 50^2 + (2 \times a \times 2000)$
 $a = -0.63 \text{ m s}^{-2}$

b $v = u + at$
 $0 = 50 + (-0.63) \times t$
 $t = 80 \text{ s}$

Alternatively, you could use $s = \frac{(u + v)}{2}t$

In the following questions, assume that all accelerations are uniform.

- 20 A ball rolling down a slope passes a point P with a velocity of 1.2 m s^{-1} . A short time later it passes point Q with a velocity of 2.6 m s^{-1} .
- a What was its average velocity between P and Q?
 - b If it took 1.4 s to go from P to Q, determine the distance PQ.
 - c Calculate the acceleration of the ball.

- 21 An aircraft accelerates from rest along a runway and takes off with a velocity of 86.0 m s^{-1} . Its acceleration during this time is 2.40 m s^{-2} .
- a Calculate the distance along the runway that the aircraft needs to travel before take-off.
 - b Predict how long after starting its acceleration the aircraft takes off.